

# Physics 214/224, Quiz 4

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04/05/2024

For grading purposes

|           |  |
|-----------|--|
| Problem 1 |  |
| Problem 2 |  |
| Problem 3 |  |
| Problem 4 |  |
| Studio    |  |
| Total:    |  |

WID:\_\_\_\_\_

NAME:\_\_\_\_\_

**Instructions.** Print and sign your name on this quiz and on your scantron card. In doing so, you are acknowledging the KSU Honor Code: “On my honor as a student I have neither given nor received unauthorized aid on this assignment.”

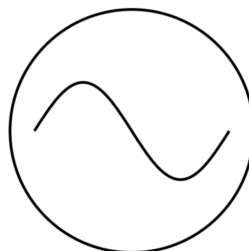
**For two quick points, circle your studio instructor:**

Lohman (TU 7:30)    Maravin (TU 9:30)    Maravin (TU 11:30)

Han (WF 9:30)

Work alone and answer all questions. Part I questions must be answered on the Scantron cards. Put your name on your card. Color in the correct box completely for every answer with a pencil. If you make a mistake, erase thoroughly. **Don't forget to color in the boxes for your WID. If we have to correct this by hand we may take off five points from your score!** Part II must be answered in the space provided on the test. The last page is a detachable equation sheet. You may use a calculator. You may ask the proctors questions of clarification. Show your work in a clear and organized manner. You may receive partial credit for partial solutions. Solutions lacking supporting calculations will receive no points.

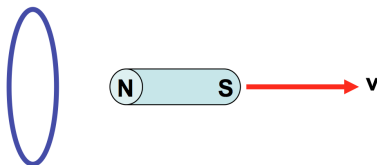
Dr. Maravin's deep thoughts in the middle of a heat wave in Europe:



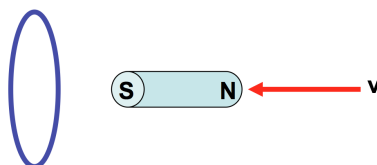
**THE AC I GET**

Part I: All questions to be answered on your Scantron card (48 points total)

1. (3 points) Which way, if any, will the current flow in the loop as the magnet is pulled out as seen from the left?



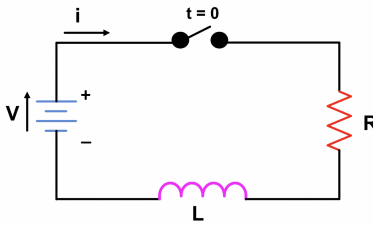
- (a) Clockwise
  - (b) Counterclockwise
  - (c) No current will flow.
2. (3 points) Which way, if any, will the current flow in the loop as the magnet is pushed in as seen from the left?



- (a) Clockwise
  - (b) Counterclockwise
  - (c) No current will flow.
3. (3 points) The quantity  $\Phi_B = \int \vec{B} \cdot d\vec{A}$  is called
- (a) Magnetic flux.
  - (b) Magnetic potential.
  - (c) Magnetic variance.
  - (d) Magnetic thingy.
4. (3 points) The integration of  $\Phi_B = \int \vec{B} \cdot d\vec{A}$  used for Faraday's law of induction is done on
- (a) over a volume.
  - (b) over a surface.
  - (c) over a closed loop.

5. (3 points) You move a strong magnet inside a ring made of wood (or any non-conductive material). Would an electromotive force (EMF) be induced around the wooden ring?
- (a) Yes.
  - (b) No.
  - (c) It depends on the material.
  - (d) It depends on how fast the magnet is being moved.

6. (3 points) In the  $RL$  circuit given below the switch is closed at  $t = 0$ . The voltage of the battery  $V = 10 \text{ V}$ , the resistance  $R = 100 \Omega$ , and the inductance  $L = 1 \text{ H}$ . What is the current  $i$  in the circuit at time  $t = 0$ ?

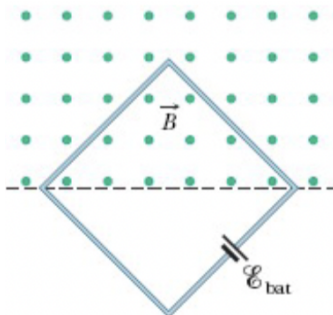


- (a) 0.1 A
  - (b) 0.0 A
  - (c) 10 A
  - (d) 0.1 V
7. (3 points) In the  $RL$  circuit from the previous problem, what is the current in the circuit at  $t = 34985$  seconds?
- (a) 0.1 A
  - (b) 0.0 A
  - (c) 10 A
  - (d) 0.1 V
8. (3 points) A toroid has  $L = 1 \text{ mH}$  inductance. You filled the cavity of the toroid with a ferrous core that has relative permeability  $\kappa_M = 1000$  and doubled the number of turns of wire of the toroid. The inductance of a re-worked toroid is
- (a) 1 mH.
  - (b) 1 H.
  - (c) 2 H.
  - (d) 4 H.
9. (3 points) The power outlet has a marking 110 V. What is the amplitude of the voltage in this AC source is
- (a) 110 V.
  - (b) 156 V.
  - (c) 78 V.
  - (d) 50 Hz.

10. (3 points) Which statement is most correct?
- (a) Inductors are short circuits at low frequencies and open circuits at high frequencies.
  - (b) Inductors are open circuits at low frequencies and short circuits at high frequencies.
  - (c) Inductors are short circuits at low and high frequencies.
  - (d) Inductors are open circuits at low and high frequencies.
11. (3 points) Which statement is most correct?
- (a) Capacitors are short circuits at low frequencies and open circuits at high frequencies.
  - (b) Capacitors are open circuits at low frequencies and short circuits at high frequencies.
  - (c) Capacitors are short circuits at low and high frequencies.
  - (d) Capacitors are open circuits at low and high frequencies.
12. (3 points) You power an ideal inductor with  $L = 1$  H by a power outlet that has  $V_{\text{RMS}} = 110$  V and  $f = 50$  Hz. What is the RMS current through the inductor?
- (a) 1 A
  - (b) 0.35 A
  - (c) 0.50 A
  - (d) 0.25 A
13. (3 points) For the circuit in the previous question, what is the average power that the inductor draws?
- (a) 0 W
  - (b) 38.5 W
  - (c) 55 W
  - (d) 27.5 W
14. (3 points) You plugged an unknown circuit into an AC voltage supply with a variable frequency. You noticed that the current drops as you increase the frequency. The circuit reactance is mostly
- (a) resistive.
  - (b) inductive.
  - (c) capacitive.
  - (d) made by aliens.

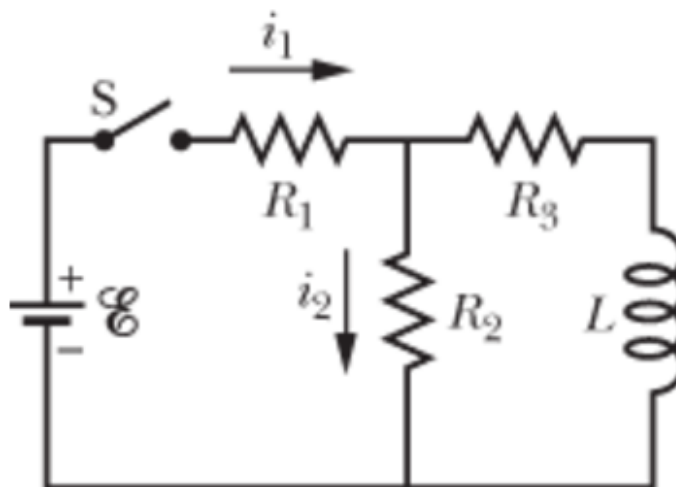
15. (3 points) While building your radio receiver, you are making a resonant circuit with an inductance of  $2.5\ \mu\text{H}$ . To be able to tune in to 97.9 MHz radio station, you need a capacitance of
- (a) 3.12 mF.
  - (b) 1.06 pF.
  - (c) 2.12 nF.
  - (d)  $5.14\ \mu\text{F}$ .
  - (e) 42.
16. (3 points) At least two transformers operate between us in Manhattan and the Jeffrey Energy Center in Bellevue, where our electric power originates. What kind of transformers are these?
- (a) Jeffrey: step-up. Manhattan: step-up.
  - (b) Jeffrey: step-up. Manhattan: step-down.
  - (c) Jeffrey: step-down. Manhattan: step-down.
  - (d) Jeffrey: step-down. Manhattan: step-up.

Part II: Work out this part in the space provided. Show your work!  
(50 points total)



17. (14 points) The magnetic flux through the square loop shown in the figure above has the following time dependence  $\Phi_B = a + bt^2$ , where  $a = 0.1 \text{ Wb}$ ,  $b = 0.15 \text{ Wb/s}^2$ , and  $t$  is given in seconds. The size of the square's side is 2.5 cm. The EMF of a battery is 3 V. Assume the magnetic field  $\vec{B}$  to be uniform.
- (a) (3 points) What is the magnitude of the magnetic field at a time  $t = 1 \text{ s}$ ?
- (b) (3 points) What is the **induced** EMF in the circuit at  $t = 1 \text{ s}$ ?
- (c) (3 points) What is the **net** EMF in the circuit at  $t = 1 \text{ s}$ ?
- (d) (2 points) What is the direction of the current in the circle at  $t = 1 \text{ s}$  (clockwise? counter-clockwise? no current?)
- (e) (3 points) Find a time when there is no current flowing through the battery.

Extra paper for calculation, notes, poetry, and reflection.



18. (12 points) In the circuit above  $\mathcal{E} = 60 \text{ V}$ ,  $R_1 = 20 \, \Omega$ ,  $R_2 = R_3 = 40 \, \Omega$ , and  $L = 3.0 \text{ H}$ . Assume positive currents  $i_1$  and  $i_2$  in the directions indicated in the figure. If the current flows in the opposite direction than what is specified, indicate it as a negative value.

(a) (4 points) Immediately after switch  $S$  is closed, what are the currents  $i_1$  and  $i_2$ ?

(b) (4 points) A long time after, what are the currents  $i_1$  and  $i_2$ ?

(c) (4 points) The switch is then re-opened. Just then, what are  $i_1$  and  $i_2$ ?



Extra paper for calculation, notes, poetry, and reflection.

19. (12 points) A 20 cm long air-core solenoid consists of 10,000 turns of wire wrapped around a  $2 \text{ cm}^2$  cross-sectional area. The solenoid is connected in series to a  $0.1 \mu\text{F}$  capacitor and a  $20 \Omega$  resistor. The resultant  $RLC$  circuit is connected to an AC power supply providing voltage with an amplitude of 400 V at a frequency that can be varied.

(a) (1 point) Draw schematics of the circuit.

(b) (4 points) Calculate the inductance of the inductor.

(c) (3 points) What operational frequency  $f_0$  of the AC power supply will result in the maximal heating of the resistor?

(d) (4 points) Calculate the average heat dissipation on the resistor at  $f_0$  that you found in the previous question.

Extra paper for calculation, notes, poetry, and reflection.

20. (10 points) A toy requires 9 V of AC voltage at 50 Hz. Unfortunately, the power supply is broken and you need to build a replacement using the following items (use only what you need, some of the items might not be necessary): several identical open-core solenoids, each having  $N = 10,000$  turns, a long iron core that allows connecting magnetic flux between the abovementioned solenoids, 10 D-type batteries (1.5 V each), connecting wires, switches, a power supply that provides 27 V AC voltage at 50 Hz, a bunch of assorted resistors, and a very old slice of pepperoni pizza.

(a) (2 points) The first step is to identify the primary source of power. Would D-type batteries be a good choice to produce the power supply for the toy? Or would a power supply be better?

(b) (6 points) Propose a circuit that would provide a 9 V AC voltage to the toy. Draw the diagram.

(c) (2 points) If the toy needs 15 W to operate, estimate the current that flows through the primary source of power when the toy is running.

Extra paper for calculation, notes, poetry, and reflection.

## Equations and Constants for Quiz 4

Note: you may detach this sheet from the quiz. You do not need to hand it in.

### 1. Point charge electric field

- (a)  $\vec{E}_Q(\vec{r}) = \frac{kQ}{|\vec{r}-\vec{r}_Q|^2} \frac{\vec{r}-\vec{r}_Q}{|\vec{r}-\vec{r}_Q|}$ ;
- (b)  $k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ ;
- (c) proton charge:  $e = 1.6 \times 10^{-19} \text{ C}$ .

### 2. Electric force: $\vec{F}_{1 \rightarrow 2} = Q_2 \vec{E}_{Q_1}(\vec{r}_{Q_2})$ ;

- (a) Coulomb's Law:  $\vec{F}_{1 \rightarrow 2} = \frac{kQ_1Q_2}{|\vec{r}_2-\vec{r}_1|^2} \frac{\vec{r}_2-\vec{r}_1}{|\vec{r}_2-\vec{r}_1|}$ ;

### 3. Superposition: $\vec{E}_{\text{TOT}}(\vec{r}) = \sum_n \frac{kQ_n}{|\vec{r}-\vec{r}_n|^2} \frac{\vec{r}-\vec{r}_n}{|\vec{r}-\vec{r}_n|}$ or $\int \frac{k dq(\vec{r}')}{|\vec{r}-\vec{r}'|^2} \frac{\vec{r}-\vec{r}'}{|\vec{r}-\vec{r}'|}$ .

### 4. Gauss's Law

- (a)  $\oint \vec{E} \cdot d\vec{A} = \frac{1}{\epsilon_0} Q_{\text{enc}}$ ;
- (b) spheres:  $4\pi r^2 E = \frac{1}{\epsilon_0} \int_0^r 4\pi r'^2 \rho(r') dr'$ ;
- (c) cylinders:  $2\pi r L E = \frac{1}{\epsilon_0} \int_0^r 2\pi r' L \rho(r') dr'$ ;
- (d) conductors:
  - i.  $E = Q = 0$  inside.
  - ii. On surface,  $\vec{E}_{\parallel} = 0$ ;  $\vec{E}_{\perp} = \sigma/\epsilon$ .

### 5. Potential/Voltage and Potential Energy

- (a)  $V_2 - V_1 = - \int_1^2 \vec{E} \cdot d\vec{r}$ ;
- (b)  $V = \sum_n \frac{Q_n}{4\pi\epsilon_0 |\vec{r}-\vec{r}_n|}$  or  $\int \frac{dq(\vec{r}')}{4\pi\epsilon_0 |\vec{r}-\vec{r}'|}$
- (c)  $PE = U = qV$ ;
- (d)  $V = \text{constant on conductor}$ .

### 6. Capacitance

- (a)  $Q = CV$
- (b)  $C_{\text{par.plate}} = \epsilon_0 A/d$
- (c)  $C_{\text{dielectric}} = \kappa C_{\text{no dielectric}}$
- (d)  $U = \frac{1}{2} CV^2 = Q^2/2C$

### 7. Current

- (a)  $I = dQ/dt$
- (b)  $\vec{J} = nq\vec{v}_{\text{drift}}$
- (c)  $J = I/A$
- (d)  $P = VI = V^2/R = I^2 R$

### 8. Resistance

- (a) Ohm's law (micro)  $\vec{J} = \sigma \vec{E}$
- (b) Ohm's law (macro)  $V = IR$
- (c)  $R = \rho L/A = L/\sigma A$
- (d)  $P = dU/dt = VI = I^2 R = V^2/R$

### 9. Circuits

- (a)  $V_b - V_a = \sum_n \mathcal{E}_n - \sum_n i_n r_n$
- (b)  $\sum_{in} I_n = \sum_{out} I_n$  (at any *junction*)
- (c)  $\sum_n \mathcal{E}_n - \sum_n i_n r_n = 0$  (around any closed *loop*)

| component | series                          | parallel               |
|-----------|---------------------------------|------------------------|
| battery   | $\mathcal{E}_1 + \mathcal{E}_2$ | — — — — —              |
| resistor  | $R_1 + R_2$                     | $(1/R_1 + 1/R_2)^{-1}$ |
| capacitor | $(1/C_1 + 1/C_2)^{-1}$          | $C_1 + C_2$            |
| inductor  | $L_1 + L_2$                     | $(1/L_1 + 1/L_2)^{-1}$ |

### (e) RC circuits:

- i.  $\tau_{\text{RC}} = RC$ .
- ii.  $Q_{\text{charging}} = Q_{\text{max}} (1 - e^{-t/\tau_{\text{RC}}})$ .
- iii.  $Q_{\text{dis-charging}} = Q_{\text{max}} e^{-t/\tau_{\text{RC}}}$ .

### 10. Magnetic force/torque.

- (a)  $\vec{F}_M = q\vec{v} \times \vec{B}$ .
- (b)  $\vec{v} \parallel \vec{B} \rightarrow$  no deflection.
- (c)  $\vec{v} \perp \vec{B} \rightarrow$  circle with  $R = \frac{mv}{qB}$ .
- (d)  $\vec{F} = I\vec{L} \times \vec{B}$ .
- (e)  $|\vec{\mu}| = IA$ .
- (f)  $\vec{\tau} = \vec{\mu} \times \vec{B}$ .
- (g)  $PE = -\vec{\mu} \cdot \vec{B}$

### 11. Magnetic fields

- (a) Biot-Savart law:  $d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{\ell} \times \vec{r}}{|\vec{r}|^3}$ .
- (b) Ampere's Law:  $\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{\text{enc.}}$
- (c)  $\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$ .
- (d)  $B_{\text{wire}} = \mu_0 I/2\pi r$ .
- (e)  $B_{\text{arc}} = \mu_0 I\phi/4\pi R$
- (f)  $B_{\text{circle}} = \frac{\mu_0 \mu}{2\pi(z^2 + R^2)^{3/2}}$
- (g)  $B_{\text{solenoid}} = \mu_0 nI$ .
- (h) Magnetism I:  $\vec{B}_{\text{TOT}} = \vec{B}_{\text{APP}} + \vec{B}_{\text{MAG}}$ .
- (i) Magnetism II:  $\vec{B}_{\text{TOT}} = K_M \vec{B}_{\text{APP}} = (1 + \chi_M) \vec{B}_{\text{APP}}$ .

12. Faraday's Law

- (a)  $\mathcal{E} = \oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A}$ ;  
 (b) Lenz's law: induced current oppose flux changes.

13. Inductance

- (a) Self:  $\Phi_B = LI$ .  
 i. solenoid:  $L = \mu_0 N^2 A / \ell$ .  
 (b) Mutual:  $\Phi_B^{(2)} = M_{12} I_1$ .  
 i. two solenoids:  $M_{12} = \mu_0 N_1 N_2 A / \ell$ .  
 (c) Energy storage:  $U = \frac{1}{2} L I^2$ .  
 (d) LR circuits:  
 i.  $I_{\text{energizing}} = I_{\text{max}} (1 - \exp(-t/\tau_{RL}))$ .  
 ii.  $I_{\text{de-energizing}} = I_{\text{max}} \exp(-t/\tau_{RL})$ .  
 iii.  $\tau_{RL} = L/R$ .  
 (e) Ideal transformer:  $\mathcal{E}_2/\mathcal{E}_1 = I_1/I_2 = N_2/N_1$ .  
 (f) AC circuits  
 i.  $\mathcal{E}(t) = \mathcal{E}_0 \cos \omega t$ .  
 ii.  $\mathcal{E}_{rms} = \sqrt{\langle \mathcal{E}^2(t) \rangle} = \mathcal{E}_0 / \sqrt{2}$ .  
 iii.  $\langle P \rangle = \mathcal{E}_{rms} I_{rms} \cos \phi$ .  
 (g) LC Oscillations  
 i.  $\omega = 2\pi f = \frac{1}{\sqrt{LC}}$   
 (h) Series LCR circuit.  
 i.  $I(t) = \frac{\mathcal{E}_0 \cos(\omega t - \phi)}{\sqrt{(X_L - X_C)^2 + R^2}}$  =  
 ii.  $X_L = 2\pi f L$ .  
 iii.  $X_C = 1/(2\pi f C)$ .  
 iv.  $\tan \phi = (X_L - X_C)/R$ .  
 (i) Damping (LCR circuit)  
 i.  $\omega_0 \rightarrow \omega' = \sqrt{\omega_0^2 - \frac{1}{4\tau^2}}$ ,  $\tau = L/R$   
 ii. if  $1/(2\tau) > \omega_0$ , system is "over-damped", no oscillations are possible  
 iii. if  $1/(2\tau) = \omega_0$ , system is critically damped

14. velocity:

- (a)  $\vec{v}_{AVG} = (\vec{r}_2 - \vec{r}_1) / (t_2 - t_1) \rightarrow \vec{v} = d\vec{r}/dt$  as  $t_2 \rightarrow t_1$ ;  
 (b) speed:  $v = |\vec{v}|$ .

15. acceleration:

- (a)  $\vec{a}_{AVG} = (\vec{v}_2 - \vec{v}_1) / (t_2 - t_1) \rightarrow \vec{a} = d\vec{v}/dt$  as  $t_2 \rightarrow t_1$ ;

(b) constant acceleration:

- i.  $\vec{v} = \vec{v}_0 + \vec{a}t$ ;  
 ii.  $\vec{r} = \vec{r}_0 + \vec{v}_0 t + \vec{a}t^2/2$ .

(c) Free fall:

- i.  $\vec{a} = 0\hat{i} - g\hat{j}$ ;  $g = 9.8 \text{ m/s}^2$ .

16. Newton's Laws

- (a)  $\vec{v} = \text{constant} \Leftrightarrow \vec{F}_{NET} = 0$ ;  
 (b)  $m\vec{a} = \vec{F}_{NET}$ ;  
 (c)  $\vec{F}(1 \text{ on } 2) = -\vec{F}(2 \text{ on } 1)$ .

17. Everyday forces

- (a) weight on Earth:  $W = mg$ ;  
 (b) tension: always pulls, constant throughout string;  
 (c) normal force:  $\perp$  to surfaces  
 (d) friction:  $\parallel$  to surfaces, opposes motion  
 i.  $f_K = \mu_K N$ ;  
 ii.  $0 \leq f_S \leq f_S^{MAX} = \mu_S N$ ;  
 iii.  $0 \leq \mu_K \leq \mu_S$ ;  
 (e) spring:  $F = -kx$ ;

18. Uniform circular motion

- (a)  $\vec{a}_C = v^2/r$  to center of circle;  
 (b) period:  $T = 2\pi r/v$ ;

19. Universal gravitation

- (a)  $\vec{F}_{G,1 \rightarrow 2} = \frac{-GM_1 M_2}{|\vec{r}_2 - \vec{r}_1|^2} \frac{\vec{r}_2 - \vec{r}_1}{|\vec{r}_2 - \vec{r}_1|}$ ;  $G = 6.7 \times 10^{-11} \text{ N}\cdot\text{kg}^{-2}\cdot\text{m}^2$ ;  
 (b) Kepler's 3rd law:  $T^2 = \frac{4\pi^2}{GM} r^3$ .

20. Work and Energy

- (a)  $W = \int \vec{F} \cdot d\vec{r} \rightarrow \int F \cos \phi dx (1D) \rightarrow Fd \cos \phi (F \text{ constant})$ ;  
 (b)  $KE = \frac{1}{2}mv^2$ ;  
 (c)  $W = \Delta(KE) = KE_2 - KE_1$  (all forces);  
 (d)  $PE = -W$  (conservative forces);  
 (e)  $PE = mgy$  (gravity near earth surface);  
 (f)  $PE = -Gm_1 m_2 / r$  (all gravity);  
 (g)  $PE = \frac{1}{2}kx^2$  (spring);  
 (h)  $KE + PE = \text{constant}$  (conservative forces);

21. Momentum

- (a)  $\vec{P} = \sum m_n \vec{v}_n$ ;  
 (b)  $\frac{d\vec{P}}{dt} = \vec{F}_{EXT}$ ;  
 (c) impulse:  $\vec{J} = \Delta\vec{P} = \vec{F}_{EXT}\Delta t$ ;  
 (d)  $\vec{F}_{EXT} = 0 \rightarrow \vec{P} = \text{constant}$ ;
22. center of mass
- (a) few objects:  $\vec{R}_{CM} = \sum m_n \vec{r}_n / M$ ; or  $\int \vec{r} dm / M$ ;  
 (b)  $\vec{P} = M \frac{d\vec{R}_{CM}}{dt} = M \vec{V}_{CM}$ ;  
 (c)  $\vec{F}_{EXT} = 0$  and  $\vec{P} = 0 \rightarrow \vec{R}_{CM} = \text{constant}$ .
23. Two-body collisions.
- (a) elastic: conserves  $KE$ ;  
 (b) inelastic: not elastic;
24. Rotation equations
- (a) For cloning translational formulas:  
 i.  $x \rightarrow \theta$ ;  $\vec{v} \rightarrow \vec{\omega} = \frac{d\vec{\theta}}{dt}$ ;  
 ii.  $\vec{a} \rightarrow \vec{\alpha} = \frac{d\vec{\omega}}{dt}$ ;  $m \rightarrow I = \sum m_n r_n^2$ ;  
 iii.  $\vec{F} \rightarrow \vec{\tau} = \vec{r} \times \vec{F}$ ;  
 iv.  $\vec{p} \rightarrow \vec{L} = \vec{r} \times \vec{p} = I\vec{\omega}$ ;  
 (b)  $KE = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ ;  
 (c) rolling without slipping:  
 i.  $v = \omega r$ ;  
 ii.  $a = \alpha r$ .  
 (d) angular momentum conservation  
 i.  $\frac{d\vec{L}}{dt} = \vec{\tau} = I\vec{\alpha}$ ;  
 ii.  $\vec{\tau} = 0 \rightarrow \vec{L} = \text{constant}$ ;  
 iii.  $\vec{\tau} = 0 \rightarrow \sum I_n \vec{\omega}_n = \text{constant}$ .
25. Static equilibrium
- (a)  $\sum \vec{F}_n = 0$ ;  
 (b)  $\sum \vec{\tau}_n = 0$ .
26. Equilibrium and PE
- (a)  $F(x) = -d(PE)/dx$ ;  
 (b)  $d^2(PE)/dx^2 > 0 \rightarrow \text{stable}$ .
27. Fluids
- (a)  $\rho = M/V$ ;  
 (b)  $P = F/A$ ;  
 (c)  $P = P_0 + \rho_F g d$ ;
- (d)  $F_{\text{buoy}} = \rho_F g V_{\text{disp}}$  (Archimedes);  
 (e)  $\nu A = \text{constant}$  (Venturi);  
 (f)  $\frac{1}{2}\rho v^2 + P + \rho g y = \text{constant}$  (Bernoulli).
28. Thermodynamics
- (a) Ideal gas:  $PV = nRT$ ;  
 (b) Ideal gas:  $\langle KE \rangle = 3kT/2$ .  
 (c) Ideal gas:  $v_{RMS} = \sqrt{3kT/m}$ .  
 (d)  $\Delta L/L = \alpha \Delta T$  (thermal expansion);  
 (e)  $\Delta Q = Mc\Delta T$  (specific heat);  
 (f)  $\Delta Q = ML$  (latent heat, to change phase).  
 (g)  $\frac{dQ}{dt} = \frac{\kappa A(T_{HI} - T_{LO})}{L}$ . (heat conduction)  
 (h)  $\frac{dQ}{dt} = \sigma \epsilon A (T_{HI}^4 - T_{LO}^4)$ . (thermal radiation)  
 (i) 0<sup>th</sup> Law: eventually,  $T_1 \rightleftharpoons T_2$ ;  
 (j) 1<sup>st</sup> Law:  $\Delta U = \Delta Q - \Delta W$ .  
 (k) 2<sup>nd</sup> Law (mechanic's version):  $\epsilon = \frac{W_{OUT}}{Q_{IN}} \leq 1 - \frac{T_{LO}}{T_{HI}}$  (heat engine).  
 (l) 2<sup>nd</sup> Law (entropy version):  $\Delta S = \frac{\Delta Q}{T} \geq 0$  (isolated system).  
 (m) Random walk:  $d_{RMS} \propto \sqrt{t}$ .